TIC 219765539

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

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BLS EPHEMERIS
  Period       0.690436 d  (ALIASED, see below)
    -> corrected period: 1.380873 d
  Epoch (t0)   2581.5406 BTJD
  Duration     0.1200 d
  Depth        0.032 %  (323 ppm)
  SNR          17.7

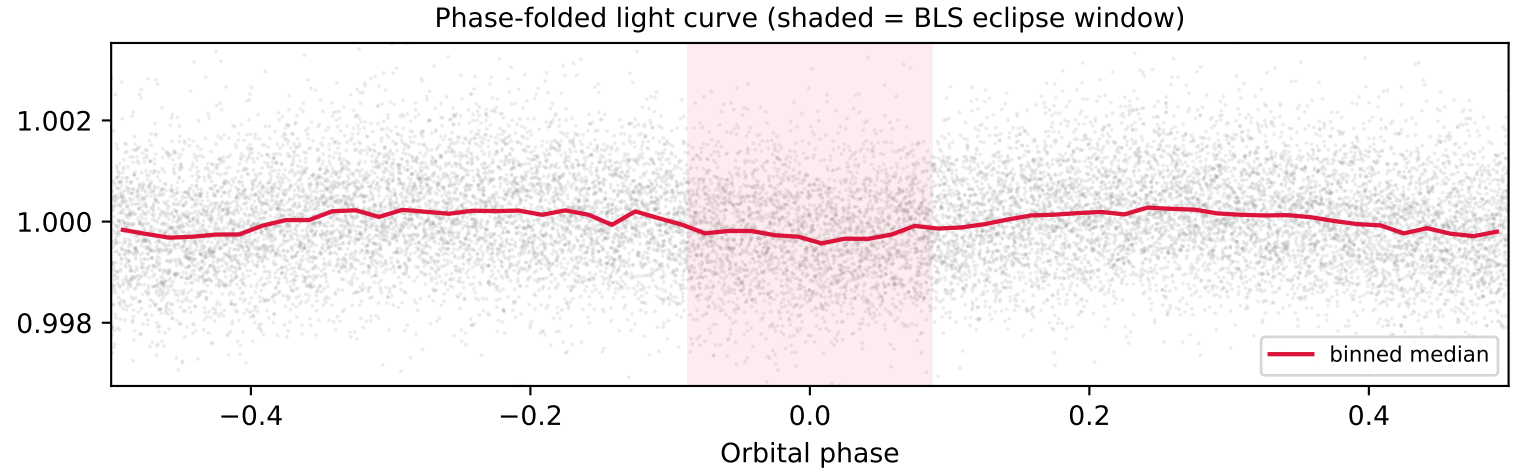
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     34  (gate >= 3)
  Robust depth ratio    0.92  (gate >= 0.50)
  Robust depth sigma    13.1  (gate >= 7.0)
  Odd/even z-score      3.64  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

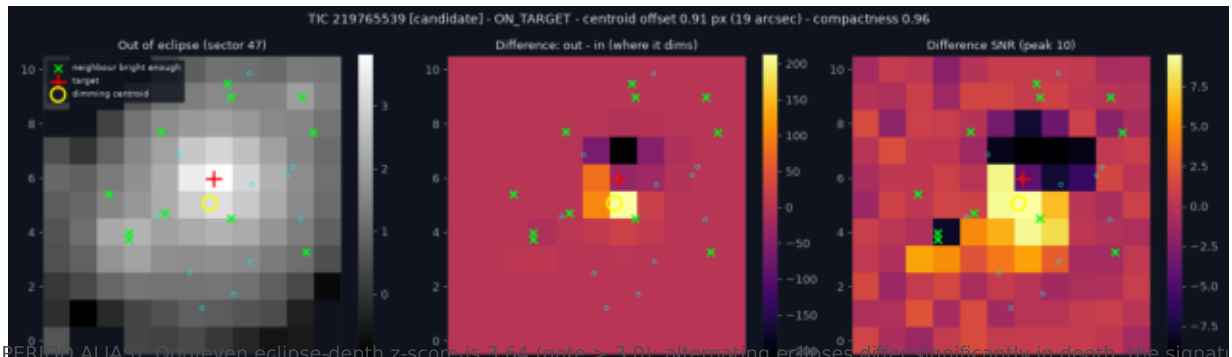
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 47    light-curve dip in this sector: 18.2 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   9.57   (gate >= 5.0)      Compactness   0.96   (gate >= 0.50)
  Dimming centroid offset     0.91 px (19 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 1400955883
    0.94 px away, dTmag +8.43 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.64 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.380873 d, in any write-up.
Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.035 px (dimming centroid 0.906 px from target vs 0.941 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.